RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Friday, 7 August 2026, 9:37 AM
State	Finished
Completed on	Friday, 14 August 2026, 9:04 AM
Time taken	6 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void reverse(int n) {
    int rev = 0, remainder;
    int count = 0;

    while (n != 0) {
        count++; // Count for while condition check (n != 0)
        count++; // Count for remainder calculation statement
        remainder = n % 10;

        count++; // Count for rev update statement
        rev = rev * 10 + remainder;

        count++; // Count for n division statement
        n /= 10;
    }
    count++; // Count for final false condition check when the loop terminates

    printf("%d", count+2);
}

int main() {
    int n;
    scanf("%d", &n);
    reverse(n);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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